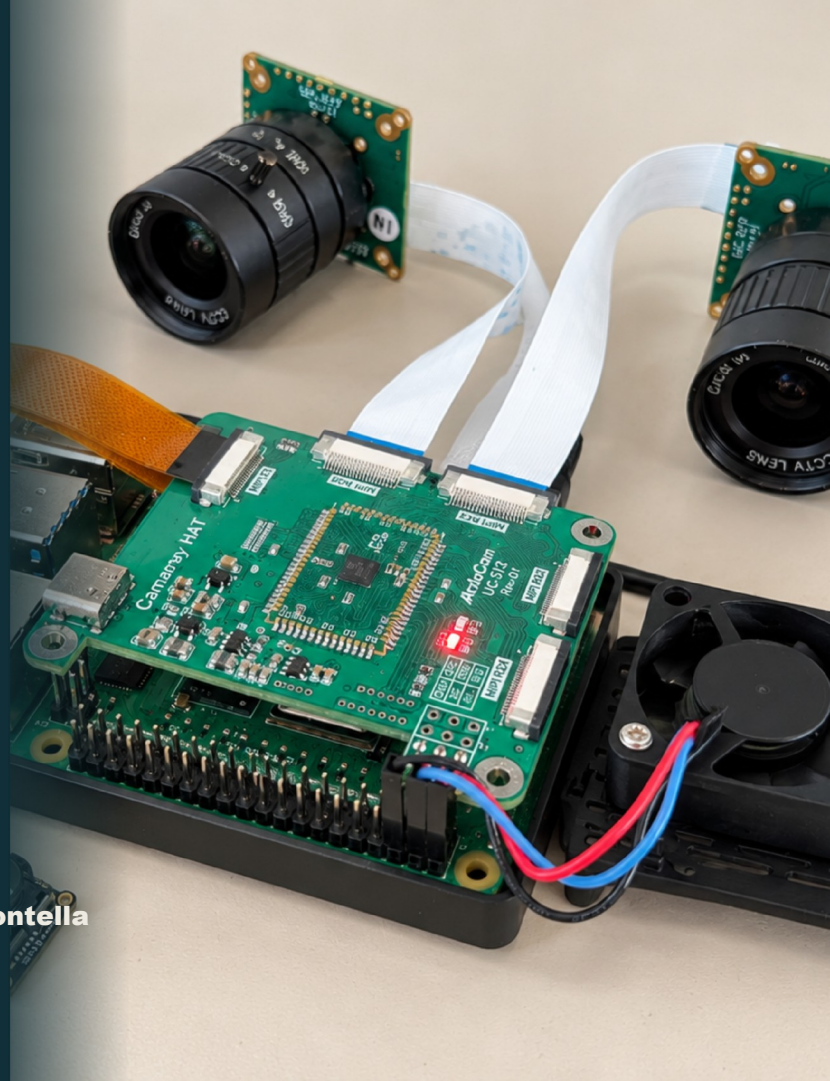


An Edge Node for Citizen-Contributed Maritime Observations

A Raspberry Pi edge prototype for concurrent RGB and thermal acquisition, offline inference and traceable maritime observation records.

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01

Problem and opportunity

Complementing established monitoring systems with repeatable close-range observations.

Coastal monitoring today

Existing systems cover different parts of the problem. Frequent close-range images during ordinary coastal activity remain scarce.

	WHAT EXISTING SOURCES PROVIDE	WHAT THEY DO NOT PROVIDE
01	Fixed stations Continuous observation at known locations	Coverage remains tied to one site
02	Satellites Wide-area coverage	Periodic passes can miss small objects near the coast
03	AIS Position and movement of equipped vessels	No image of the surrounding environment
04	Research vessels Detailed scientific observations	Data collection follows dedicated campaigns

EASY investigates a complementary layer: small, low-cost nodes that collect repeated close-range images during normal coastal activity.

Recreational boats, fishers and coastal communities may contribute observations, provided that each image keeps its acquisition context.

Citizen-contributed maritime observations

The idea in four steps

01 Complement existing monitoring

Low-cost devices add close-range observations during ordinary coastal activity.

02 People who already observe the sea

Recreational sailors, fishers and coastal communities regularly see the same areas.

03 Devices where observations happen

The node can operate on a vessel or at a fixed coastal location.

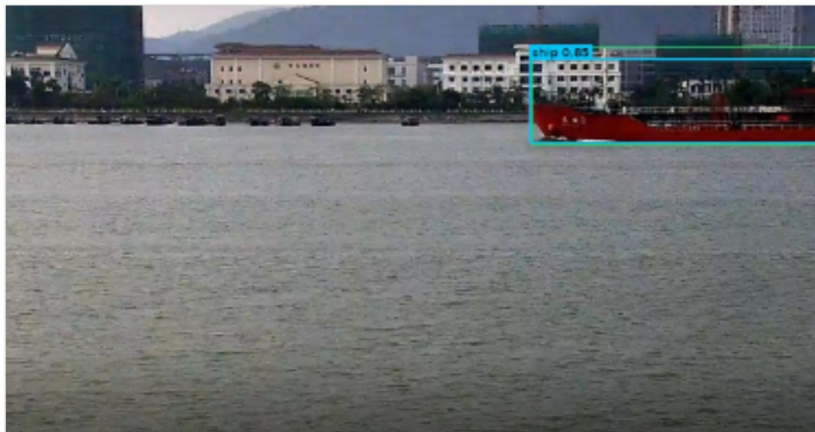
04 Make every observation traceable and repeatable

For scientific reuse, the record must explain how the data was acquired, how it was processed and under which conditions it can be shared.

This is the role of the edge node developed in our work.

From a detection to an observation record

A bounding box alone is not enough for scientific reuse.



Detected vessel in an input frame

MODEL OUTPUT

What the detector sees

- **Object class** vessel
- **Bounding box** where it appears
- **Confidence** how certain the model is

OBSERVATION CONTEXT

What makes the result reusable

- | | |
|-------------------------|--------------------|
| Capture time and source | Device state |
| Model and configuration | Processing history |

A traceable observation record

The result and its acquisition context can be reviewed and repeated later.

02

System design

Hardware, software architecture and the observation record that make the edge node measurable and reusable.

Research questions

We test the workflow, its performance on the Raspberry Pi and the information an edge node must share.

01

SOURCE PATH

Can live camera feeds and recorded frames use the same model-independent workflow?

Both sources enter through the same interface.

02

RESOURCES

Can the full stack run on a Raspberry Pi 4 with measured resource use and latency?

We measure the full service, not inference alone.

03

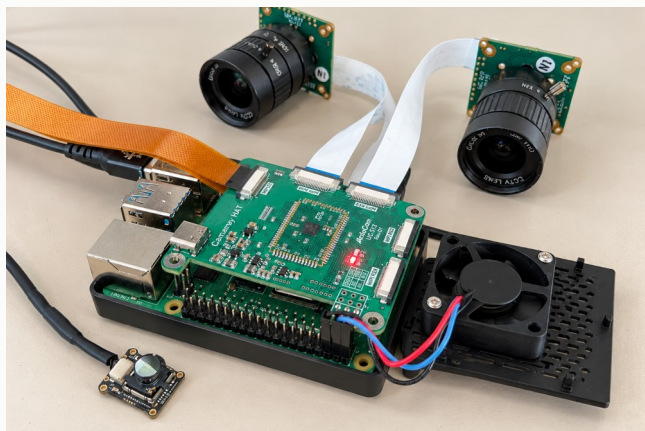
OBSERVATORY

What interfaces and metadata does an edge node need to contribute to an observatory?

The node must expose its records and sharing conditions.

From laboratory prototype to integrated node

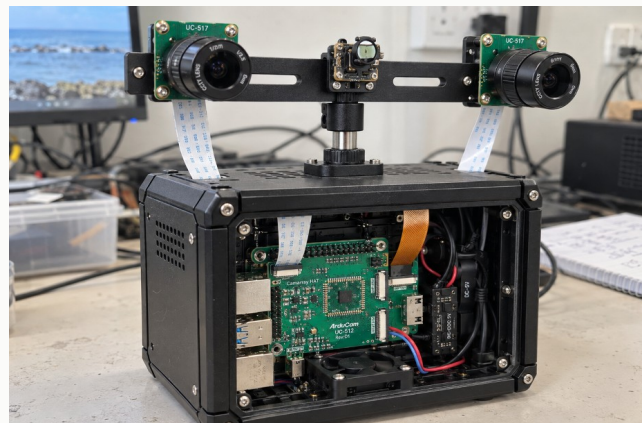
The system on the left is the hardware used for the measurements. The enclosure on the right is a mechanical integration target.



CURRENT LABORATORY PROTOTYPE, MEASURED

Raspberry Pi 4, two RGB cameras and FLIR

Lepton ONNX on CPU, active cooling, offline capture and storage



TARGET INTEGRATED NODE, DESIGN STUDY

Rigid sensor mount and transportable enclosure

Integration concept only; it was not used for the reported measurements.

EVIDENCE BOUNDARY

All performance, endurance and thermal results refer to the current laboratory prototype. Concurrent RGB and thermal acquisition is implemented and validated. Precise cross-sensor timestamp alignment for RGB-thermal fusion remains future work.

Three requirements for the software

After developing the hardware prototype, we defined these requirements before designing the architecture.

01

Camera-independent

The software should not depend on one particular camera.

02

Repeatable replay

Recorded frames should follow the same processing path as live capture.

03

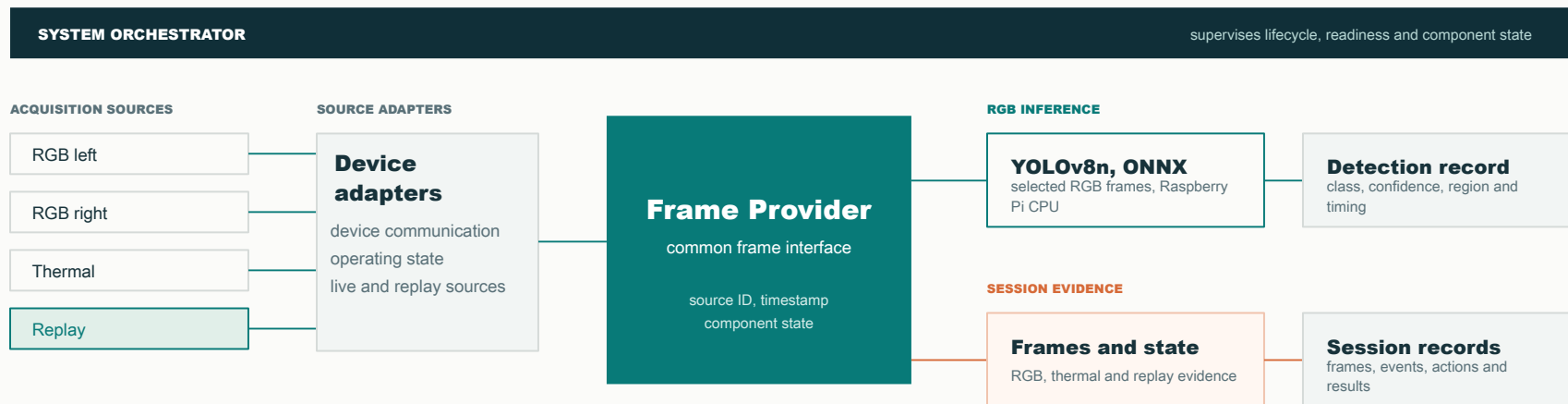
Traceable results

Every result must retain source, device state and processing context.

The next slide shows the components that implement these three requirements.

System architecture

All sources reach the Frame Provider through adapters. Selected RGB frames enter YOLOv8n, while every source can remain session evidence.



Current implementation: multimodal acquisition, RGB-only inference.

Thermal frames are stored as session evidence. RGB and thermal fusion remains future work.

What is stored with every observation

Each observation is saved together with six pieces of context.

OBSERVATION RECORD

result plus context

01

Source

which acquisition source produced it

02

Session

which collection session it belongs to

03

Capture time

when the frame was acquired

04
Device
state

how the device was operating

05

Model configuration

which model and settings were used

06

Processing times

how long each stage took

This context explains how the observation was produced and allows the processing to be repeated later.

What a record does not establish by itself

The edge node can create evidence. Scientific reuse needs an additional decision.

WHAT THE EDGE NODE AND ITS RECORD PROVIDE

Model confidence

How certain is the detector?

Record quality

Are the evidence and context complete?

WHAT REVIEW MUST DECIDE

Scientific acceptance

Can the observation be reused?

Future review and curation

A high confidence score does not automatically make an observation scientifically valid.

03

Experimental evaluation

We measure the behaviour of the complete Raspberry Pi system.

The next slide introduces the five tests used to evaluate functionality, resources, latency, thermal stability and concurrent acquisition.

Five tests, one Raspberry Pi system

Each test answers one practical question about the complete service.

01	Functional tests	Does the service work as expected?	310 automated tests
02	Steady capture	What resources does acquisition use?	300 seconds
03	Replay benchmark	What does one full request cost?	50 requests
04	CPU stress	Does the system remain thermally stable?	120 seconds
05	Concurrent endurance	Can all three sources remain active?	2 hours

The next slides focus on resources, latency, thermal stability and model validation.

Two hours of concurrent acquisition

The question was simple: can both RGB views and the thermal sensor remain active together?

YES

All three sources remained active under laboratory conditions.

RGB left

9.55 FPS

Target: 10 FPS

RGB right

9.55 FPS

Target: 10 FPS

Thermal

9.11 FPS

Target: 10 FPS

WHAT REMAINED STABLE

64,801

valid thermal frames

0

invalid thermal frames

0

lost RGB views

0

service restarts

This validates concurrent acquisition. It does not validate temporal alignment or RGB-thermal fusion.

Can steady capture run comfortably on the Pi?

Yes. Acquisition leaves most CPU and memory capacity available.

53%

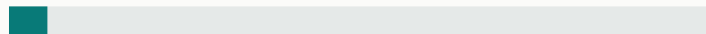
of one CPU core



steady capture

450 MiB

of 8 GiB RAM



steady capture

56 °C

mean CPU temperature

0

throttling events

What changes during inference?

The next slide measures the full request latency.

Where one second of processing time goes

Fifty replay requests measured the complete path from input frame to stored result.

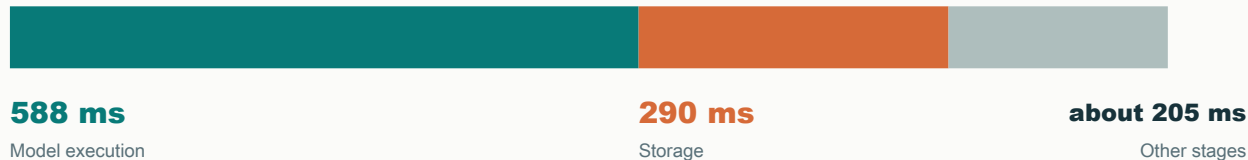
1.08 s

mean end-to-end latency

0.94 s median

across 50 requests

MEAN LATENCY BREAKDOWN



Model execution is the largest component, but storage also contributes substantially.

How to read this result

The benchmark measures one complete request. It does not mean that detection runs continuously at the camera frame rate.

Thermal stability under stress

The device remained stable during laboratory stress and endurance tests.

120 s

CPU stress

all four cores at full load

77.4 °C

maximum temperature

0.6 °C below the 78 °C cutoff

0

throttling events

during stress and endurance tests

2 h

inference endurance

2,530 requests without unrecovered errors

Laboratory evidence only. With 0.6 °C of margin under full load, enclosed or sunlit deployments need dedicated cooling.

A valid split changed the model result

The first score was inflated because nearby frames from the same video appeared in training and test data.

WHAT WENT WRONG

0.942

mAP50

Frame-level split

Similar frames from the same video leaked across the dataset partitions.

Misleading result

HOW WE CORRECTED IT

Keep each complete video sequence in one partition.

Training videos

never contribute frames to the test set

Test videos

remain unseen during model training

WHAT THE VALID TEST SHOWED

0.382

valid baseline mAP50

0.627

mAP50 after adding public data

This is the model deployed on the node.

External check

MODD2 recall: 3.9%

This separate dataset uses a different metric, so it is not directly comparable. It shows that small, distant objects near the horizon remain difficult.

04

Evidence and next steps

The laboratory prototype is measured. Field deployment is the next stage.

Answers to the three research questions

The evidence is strongest for the shared pipeline and Raspberry Pi operation. Observatory participation still requires field validation.

01 Can live and recorded sources use one contract?

Validated

Yes. The same pipeline is exercised by live and replay experiments.

02 Can the complete service run on a Raspberry Pi 4?

Validated with limits

Yes, under the tested conditions. Inference speed remains a limitation.

03 What is needed to join a scientific observatory?

Future field validation

Traceable records and interfaces are defined. Contributors, consent and federation need field validation.

What this work demonstrates

A complete path from sensing to a reusable observation record on a Raspberry Pi.

01 Acquire

Two RGB views and thermal data remain active concurrently.

02 Process locally

The Raspberry Pi runs object detection on selected RGB frames without a continuous cloud connection.

03 Preserve the context

Each result retains the source, device state, model configuration and processing information needed to interpret it.

Validated in the laboratory on Raspberry Pi 4

Next step: controlled coastal deployment

Thank you.

Questions and discussion

From a laboratory prototype to controlled coastal deployment.

A Raspberry Pi edge node for concurrent RGB and thermal acquisition, local inference and traceable maritime observation records.

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Repository and materials



model and dataset



node runtime

Scan to access code, benchmarks and supporting material.

Inference stages, 50 replay requests

STAGE	MEAN (MS)	MEDIAN	P95
Frame acquisition	60.1	57.5	86.0
Preprocessing	50.0	46.6	63.5
Model backend	588.4	551.2	782.1
Postprocessing	2.5	1.6	3.6
Preview	84.1	70.5	210.5
Storage	290.0	187.5	851.3
Request pipeline	1083.8	937.8	1675.3
API wall time	1088.9	n/a	n/a

Raspberry Pi 4, fresh service and session, final paper measurements (Table VI). YOLOv8n at 640 px: boat, ship, buoy. P95 is the 48th of 50 values.

Why the numbers moved

METRIC	PAPER, AUGUST	FINAL PAPER	REASON
Process CPU	30 %	53 %	Login, snapshot database, richer diagnostics, new model
Memory	292 MiB	450 MiB	The same additions
Model backend	574 ms	588 ms	New model trained with ABOships
End-to-end	0.91 s	1.08 s, median 0.94	Backend plus a longer storage tail
Readiness	6.5 s	10.9 s	More services start at boot
Automated tests	41	310	Frontend and security tests added

Observatory interface requirements

CONCERN	CONTRACT	STATUS
Provenance	source, session, capture and processing time, model, device state	IMPLEMENTED
Repeatability	common live and replay path, stage timing	IMPLEMENTED
Data minimization	local retention, export and retention controls	PARTIAL
Privacy	consent, sensitive locations, image filtering	POLICY
Quality control	diagnostics, scores, duplicates, cross-node agreement, review	PARTIAL
Federation	enrolment, signing, secure sync, metadata export	PLANNED
Feedback	maps, summaries, explanations of outcomes	FIELD STUDY